

CSE 350

Digital Electronics and Pulse Techniques

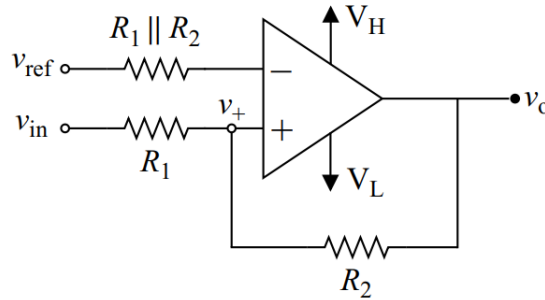
Assignment-02

Total – 20 marks

Question 1

05

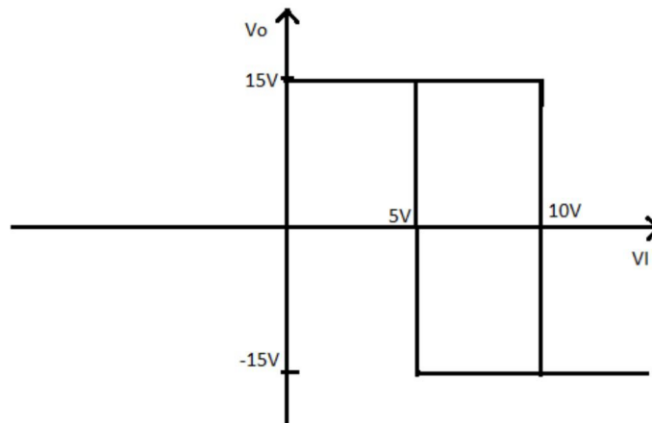
For the Given Schmitt Trigger Circuit, $V_H = 10V$, $V_L = -10V$, $V_{ref} = 1V$



(a)	Is this an Inverting or non-inverting Schmitt Trigger?	1
(b)	Let $R_1 = 1 \text{ k}\Omega$, $R_2 = 2 \text{ k}\Omega$; Draw V_{out} vs. V_{in} graph (VTC) or Transfer curve with appropriate labels.	2
(c)	Let input voltage be 18V peak to peak sine wave. Draw V_{out} vs. time plot for the given input.	2

Question 2

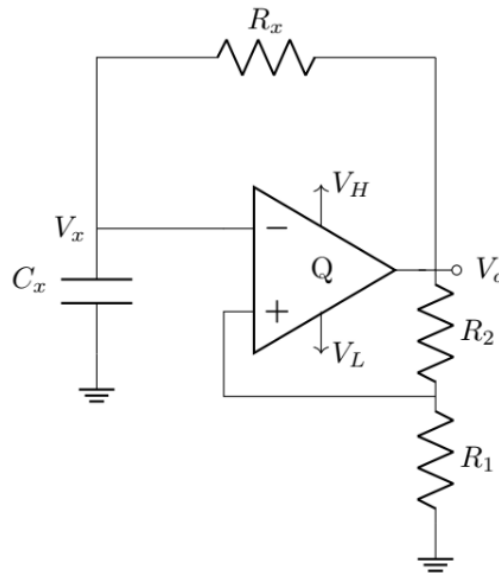
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(a)	Identify the Schmitt trigger circuit from the VTC and also find out V_H , V_L , V_{UT} , V_{LT} , V_{shift}	2.5
(b)	Design a Schmitt trigger circuit that will produce the same VTC.	2.5

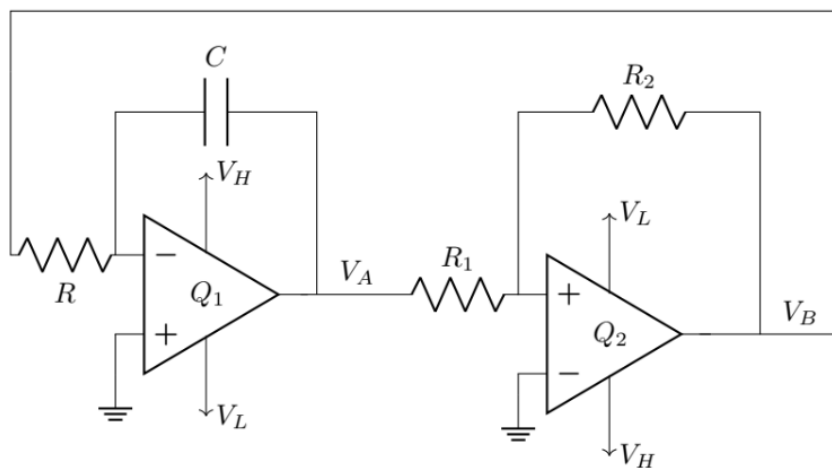
Question 3

05



The above circuit generates a square wave with 200Hz frequency and 50% duty cycle. Here, $R_1 = R_2$, $V_H = +8V$, $V_L = -8V$.

(a)	If the capacitance value is $C_X = 100nF$, Find the value of R_X and Time constant.	2
(b)	Draw the input-output waveform on the same graph with proper labeling on the amplitude and time axis	1



Here, $R_1 = 10K$ and $R_2 = 80K$

(c)	For the above circuit if $V_L = -V_H$, show that the frequency will be $f = 2/RC$ (show proper calculation)	2
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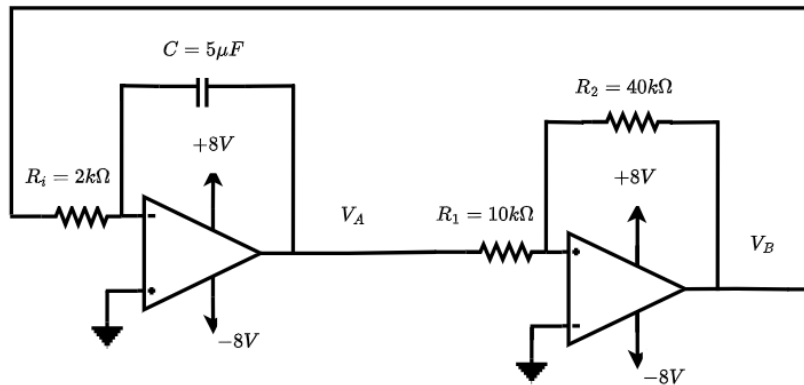


Figure 1: Triangular Wave Generator

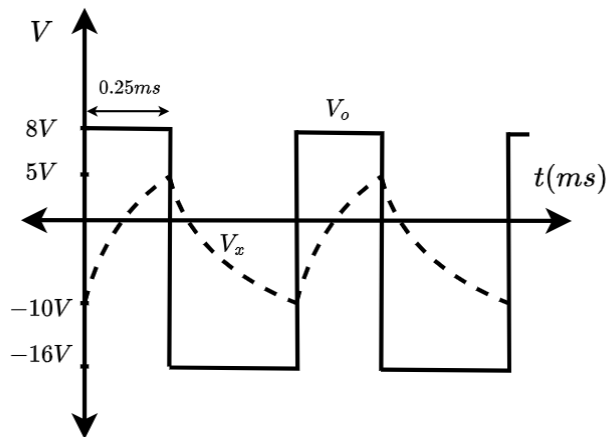


Figure 2: Square Wave Generator Output

(a)	Calculate the time period and the frequency of the triangular wave generator of Figure 1.	1
(b)	Design a square wave generator that can generate a similar output waveform given in Figure 2. Draw the designed circuit with all necessary values. (You can assume the value of the capacitor 50nF.)	2
(c)	Find the duty cycle and frequency of the output waveform of your designed square wave generator.	2